Hiten Dalmia

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Education

Chennai Mathematical Institute — Chennai, India
Master of Science in Theoretical Computer Science (On Full Scholarship) — *Aug. 2026 – Discontinued*

Kirori Mal College, University of Delhi — Delhi, India
Bachelor of Science (Honours with Research) in Mathematics, Minor in Economics — *Nov. 2022 – May 2026*

Experience

Backend Engineer — Mar. 2023 – Present
Freelance & Fractal — *Remote*

- Delivered 10+ production web and enterprise backend applications for client businesses, specializing in object-oriented Java and Node.js microservices, relational databases, RESTful APIs, and CI/CD pipelines.
- **Order Procurement & Management Solution** | *Java, Spring Boot, PostgreSQL, RESTful APIs, Docker*
  - Designed a modular enterprise order processing backend using Core Java and Spring Boot, persisting transactional data to PostgreSQL via Hibernate ORM and exposing secure RESTful APIs.
  - Created a dual-tier administration backend applying OOP design patterns for task dispatch and audit logging, establishing JUnit testing standards and leading peer code reviews in Agile sprints.
  - Developed an automated order ingestion pipeline in Java that parsed incoming unstructured messaging payloads into relational PostgreSQL records, reducing manual processing latency.
  - Optimized PostgreSQL relational schemas and Hibernate indexing constraints, tuning database queries and JPA repositories to accelerate high-volume transaction throughput.

Software Engineer — Oct. 2023 – Apr. 2025
Tagglabs Experimental Pvt. Ltd. (Contract) — *Gurugram, Haryana (Remote)*

- **Cross-platform AR App** | *Core Java, Spring Boot, C++, Flutter, RESTful APIs* **(Live link)**
  - Architected a native C/C++ bridge using FFI to port OpenCV SIFT keypoint algorithms into mobile apps, applying Object-Oriented C++ principles to encapsulate cross-platform vision logic.
  - Engineered a real-time camera feed pipeline normalizing OS frame buffers and aspect ratios, managing C memory buffers to eliminate memory leaks while maintaining 30+ FPS execution.
  - Implemented asynchronous coordinate transmission and real-time object recognition, optimizing multi-threaded data pipelines to render spatial overlays with minimal latency.
  - Built high-throughput RESTful backend APIs in Spring Boot and Dockerized microservices on AWS, integrating location-aware spatial discovery endpoints for spatial asset retrieval.
- **Kia National Conference App** | *Java, RESTful APIs, Docker, Git*
  - Constructed an event management system for 500+ attendees at the Kia National Conference, building RESTful APIs, JIRA-tracked Agile tasks, and Jenkins CI/CD deployment pipelines.

Technical Skills

Languages & Concepts: Core Java, Object-Oriented Programming (OOP), SQL, C++, Python, JavaScript
Frameworks & Databases: Spring Boot, Spring, Hibernate, PostgreSQL, MySQL, MongoDB, Node.js
APIs & Architecture: RESTful APIs, Microservices, Design Patterns, JUnit, Multithreading
DevOps & Tools: Git, Jenkins, Docker, CI/CD, AWS, Linux, Bash, JIRA, Agile Methodologies

Achievements

Erasmus Mundus Joint Master MATHS-DISC Fellowship — Europe
Issued by European Union and affiliated universities — *Mar. 2026*

Master in Mathematics (ISI) and MSc Computer Science (CMI) Scholarships — India
Indian Statistical Institute and Chennai Mathematical Institute respectively — *Jun. 2026*

Publications

Computable Analysis: Mathematical Foundations and Computability of Real Functions
Poincare Journal of Mathematical Physics & Applications, Vol. 1, No. 1 (2026), 1–20 (with Sujit Kumar)